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EDUCATION

PhD in Astronomy 05/2024 - Present

Project title: Binary systems of compact objects in the Galaxy and the resulting mass distributions;

Fellowship: FAPESP-DD I (grant number 2025/01653-5);

Institute of Astronomy, Geophysics and Atmospheric Sciences of the University of São Paulo.

Bachelor's degree in Physics 01/2020 - 12/2023

Scientific Initiation: Dark Energy influence in the Large Scale Structures formation in the universe;

Fellowship: CNPq (grant number 125181/2022-1)

Institute of Physics of the University of São Paulo.

Technical Course Integrated to Secondary Education 01/2017 - 12/2019

Technical Course: Chemistry;

ETEC Júlio de Mesquita.

PUBLICATIONS

Scientific articles

- Rocha, L., **Pires, N.**, de Sá, L., Martins, B., Barão, L., Sampaio, G., Rodrigues, D., Takahashi, R., de Avellar, M., Horvath, J., Bernardo, A. *The Galactic Black Hole mass distribution*. [doi:arXiv.2609.14154](https://arxiv.org/abs/2609.14154) (accepted for publication in ApJS);
- de Avellar, M., Horvath, J., **Pires, N. D.**, Rocha, L., de Sá, L., Martins, B., Barão, L., Rodrigues, L., Sampaio, G., Takahashi, R. *The mass of the neutron star in 4U 1820-30 revisited*. JHEAp, v. 54, 2026. [doi:10.1016/j.jheap.2026.100651](https://doi.org/10.1016/j.jheap.2026.100651);
- Benvenuto, O. G., de Vito, M. A., Echeveste, M., Novarino, M. L., **Pires, N. D.**, de Sá, L. M., Horvath, J. E. *Formation and nature of “Huntsman” binary pulsars*. A&A, v. 698, p. L5, 2025. [doi:10.1051/0004-6361/202554549](https://doi.org/10.1051/0004-6361/202554549);
- Benvenuto, O., **Pires, N.**, Horvath, J. *The effects of the irradiation feedback on the “Huntsman” binary systems* (submitted to A&A).

Proceedings

- Rocha, L. R., **Pires, N. D.**, de Sá, L., Martins, B., Sampaio, G., Rodrigues, D., Barão, L., Takahashi, R., de Avellar, M., Horvath, J., Bernardo, A. *Galactic Black Holes masses* (submitted to Astronomische Nachrichten);
- Horvath, J. E., de Sá, L. M., Rocha, L. S., **Pires, N. D.**, Miwa, R., Rodrigues, D., Barão, L. G., Chinen, G. Y. *Investigating the Evolution of Compact Object Populations*. Astronomische Nachrichten, v. 1, p. 1, 2025.

Books

- Horvath, J., Rocha, L., de Sá, L., **Pires, N. A. D.**, Avellar, M., Barão, L., Rodrigues, D., Takahashi, R., Gusmão, L., Chenin, G. *O Bê-a-bá da Astrofísica: um Guia para Desnorteados*. 1st ed. Livraria da Física, 2025.

EXPERIENCE

Institute of Astronomy, Geophysics and Atmospheric Sciences of the University of São Paulo

PhD in Astronomy

05/2024 – Present

- Project title: Binary systems of compact objects in the Galaxy and the resulting mass distributions;
- Advisor: Prof. Dr. Jorge E. Horvath;
- Fellowship: FAPESP.

Scientific Initiation

06/2021 – 02/2022

- Project title: Analysis of the X-Ray emission in different galaxies;
- Advisor: Prof. Dr. Gastão César B. Lima Neto;
- No fellowship.

Institute of Physics of the University of São Paulo

Scientific Initiation

03 – 09/2023

- Project title: Dark Energy influence in the Large Scale Structures formation in the universe;
- Advisor: Prof. Dr. Luis Raul W. Abramo;
- Fellowship: CNPq (grant number: 125181/2022-1).

Teaching Assistant

- Foundations in Astronomy 02 – 06/2026;
- Optics 02 – 07/2025;
- Physics I 08 – 12/2024;
- Physics I 08 – 12/2023;
- Electricity and Magnetism II 03 – 06/2022;
- Physics II 08 – 12/2021.

Federal University of the ABC (UFABC)

Scientific Initiation

08/2018 – 07/2019

- Project title: Natural radioactivity measurement from different types of edible salts;
- Advisor: Prof. Dr. Felipe Chen Abrego;
- Fellowship: CNPq/PIBIC-Jr (grant number: 117440/2018-3).

PARTICIPATION IN EVENTS

9th Heidelberg International Symposium on High-Energy γ -ray Astronomy

Heidelberg, Germany.

09/2026

XLVIII Reunião Anual da SAB

Hotel Glória, Caxambu, Minas Gerais - Brazil.

10/2025

1st POSYDON School

The SkAI Institute, SKAI, United States of America.

09/2025

MESA Summer School Katholieke Universiteit Leuven, KU Leuven, Belgium	06/2025
XLVII Reunião Anual da SAB Hotel Majestic, Águas de Lindoia, São Paulo - Brazil.	09/2024

PRESENTATIONS

9th Heidelberg International Symposium on High-Energy γ-ray Astronomy <i>Poster</i>	09/2026
Presentation of the Poster: “ <i>Spider Pulsars and the Intrabinary Shock</i> ”.	
RASAB 2025 <i>Poster</i>	10/2025
Presentation of the Poster: “ <i>Formation and Nature of Huntsman Binary Systems</i> ”.	
RASAB 2024 <i>Poster</i>	09/2024
Presentation of the Poster: “ <i>The Galactic BH mass distribution</i> ”.	

LANGUAGES

Portuguese (native), English (fluent), Spanish (advanced), French (basics).